

HARISH KUMAR SAMALA

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PROFESSIONAL SUMMARY

Software Engineer with experience in designing and deploying production-grade AI/ML and full-stack applications. Specialized in Generative AI, Conversational RAG pipelines, and LLM-powered systems using LangChain, OpenAI APIs, and Mistral. Skilled in prompt engineering, semantic search, and embedding-based retrieval. Strong backend foundation in Python (FastAPI, Flask, Django) with hands-on experience building scalable RESTful APIs, ETL pipelines, and optimizing database performance. AWS Certified Cloud Practitioner with CI/CD, Agile/Scrum, and cross-functional team collaboration experience.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL

Backend & Frameworks: FastAPI, Flask, Django, Streamlit, Spring Boot, SQLAlchemy, LangChain, LangGraph

Generative AI & LLMs: OpenAI API, Mistral, Groq API, LLaMA 3, HuggingFace Embeddings, Prompt Engineering, RAG, Agentic AI

Vector Databases: FAISS, Chroma, Semantic Search, RecursiveCharacterTextSplitter

Databases: PostgreSQL, MySQL, MongoDB

Data & ML: Pandas, NumPy, PySpark, Scikit-learn, ETL Pipelines, Sentiment Analysis, Data Visualization

Cloud & DevOps: AWS, Azure, GCP, Docker, CI/CD, Jenkins, Azure DevOps, GitHub, Git

Testing & Tools: JMeter, Nginx, Selenium, BeautifulSoup, Postman, Agile/Scrum

PROFESSIONAL EXPERIENCE

AI/ML Software Engineer

Jul 2024 – Present

Advanced Knowledge Technologies, NJ, USA

- Built and deployed a production Conversational RAG system (LangChain, Groq, FAISS/Chroma, Streamlit) with persistent multi-turn memory, reducing support response times by 40% and improving retrieval precision by 35%.
- Designed modular ETL pipelines for document ingestion, vector embedding, and semantic search, cutting query latency by 25%.
- Developed a domain-specific Generative AI chatbot using Mistral LLMs with history-aware prompt engineering, saving 10+ hours/week of manual operations.
- Built RESTful APIs using Flask and Django; integrated CI/CD pipelines with Azure DevOps.
- Optimized PostgreSQL and MySQL performance via indexing, caching, and query tuning, reducing execution time by 40%.
- Engineered web scraping and data pipelines (Selenium, BeautifulSoup, Pandas, PySpark) for real-time analytics.
- Applied ML techniques including Apriori algorithm and sentiment analysis to improve recommendations.
- Built dashboards using Python Dash and automated notifications using smtplib.

Python Developer

Sep 2021 – July 2022

Adithri Technologies Private Limited, Hyderabad, India

- Developed and deployed end-to-end machine learning pipelines using Python, Scikit-learn, and Azure Anomaly Detection, leveraging NumPy, Pandas, and SciPy for data preprocessing and implementing serverless workflows with AWS Lambda.
- Engineered scalable ETL pipelines using PySpark and Hive, integrating Spark with HDFS, Amazon S3, Kafka, Cassandra, and relational databases; developed RESTful microservices and APIs using Flask and Django connected to Cosmos DB, Redshift, and Azure SQL Data Warehouse.
- Designed and optimized database solutions across Oracle PL/SQL, MySQL (partitioning and sharding), DynamoDB, and SQL Server; implemented stored procedures, QuerySet optimizations, and raw SQL queries for complex data processing operations.
- Built responsive web interfaces using HTML, CSS, and JavaScript, and developed reusable Angular 12 TypeScript components utilizing NgModules, Observables, and Angular Router for seamless REST API integration.

- Managed cloud infrastructure across AWS and Azure, including CloudWatch monitoring that improved response times by 25%, S3 lifecycle policy automation, Azure Storage Explorer administration, and CI/CD pipeline implementation using Azure DevOps and GitHub for ML model deployments.

PROJECTS

SparkleClean – Full-Stack Service Booking Platform 2025

React, TypeScript, Tailwind CSS, Spring Boot, MySQL

- Built a full-stack booking platform with customer and admin portals using React and TypeScript with role-based authentication.
- Designed multi-step booking workflows with real-time status tracking and CRUD operations.
- Developed backend services using Java 17, Spring Boot, and Spring Data JPA with MySQL.
- Deployed frontend on Vercel and backend on Railway with production-grade configuration.

Superior Limousine LLC – Corporate Website 2025

TypeScript, Vite, Vercel, Cloudinary

- Built a fully responsive corporate website with a mobile-first and accessibility-focused design.
- Implemented SEO-optimized structure for improved search visibility.
- Integrated Framer Motion animations and Cloudinary media delivery.
- Deployed on Vercel with DNS configuration.

Case Notes System – NJIT 2024

NestJS, TypeScript, Prisma, PlanetScale, AWS Lambda

- Developed a hospital patient case document summarization system.
- Built scalable APIs using NestJS and Prisma ORM.
- Integrated AWS Lambda for serverless processing.
- Tested APIs using Postman.

Student Performance Prediction – NJIT 2024

Python, Scikit-learn, Pandas, NumPy, Matplotlib

- Built machine learning models using Scikit-learn to predict student performance.
- Performed data preprocessing and analysis using Pandas and NumPy.
- Visualized trends and insights using Matplotlib.

EDUCATION

Master of Science, Computer Science May 2024

New Jersey Institute of Technology (NJIT), Newark, NJ

Bachelor of Engineering, Computer Science Aug 2021

Siddhartha Institute of Technology (SITS), Hyderabad, India

CERTIFICATIONS

- AWS Certified Cloud Practitioner
- Generative AI Foundations
- Data Structures and Algorithms with Python